

Unemployment and Inflation

ECON 101 – Principles of Macroeconomics

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- 5.1 Measuring the Unemployment Rate and the Labour Force Participation Rate
- 5.2 Types of Unemployment
- 5.3 Explaining Unemployment
- 5.4 Measuring Inflation
- 5.5 Using Price Indexes to Adjust for the Effects of Inflation
- 5.6 Real versus Nominal Interest Rates
- 5.7 Does Inflation Impose Costs on the Economy?

Why Unemployment and Inflation Matter

- Previous chapter: how to measure **total output** (GDP).
- This chapter: how to measure
 - **Unemployment** in the labour market, and
 - **Inflation** in the goods and services market.
- **Canadian context:** In 2024, the unemployment rate averaged $\approx 6.3\%$ and CPI inflation was $\approx 2.7\%$ — both central targets of Bank of Canada policy (Bank of Canada MPR, 2024).

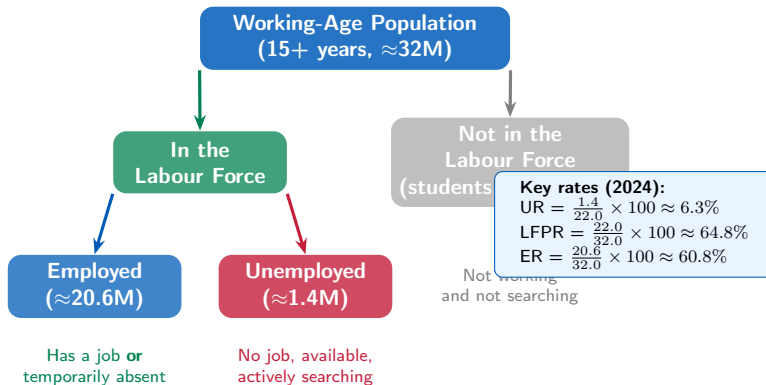
5.1 Learning Objective

Goal

Understand how unemployment, labour force participation, and employment rates are measured.

- Canada has ≈ 32 million working-age people — we cannot track everyone individually.
- Statistics Canada uses the **Labour Force Survey (LFS)**: $\approx 56,000$ households sampled monthly.
- Each working-age person (15+) is classified as **employed**, **unemployed**, or **not in the labour force**.

Figure 5.1 – Employment Status: How the LFS Classifies Canadians



Key Labour Market Rates

- Unemployment rate (UR):

$$UR = \frac{\text{Number unemployed}}{\text{Labour force}} \times 100$$

- Labour force participation rate (LFPR):

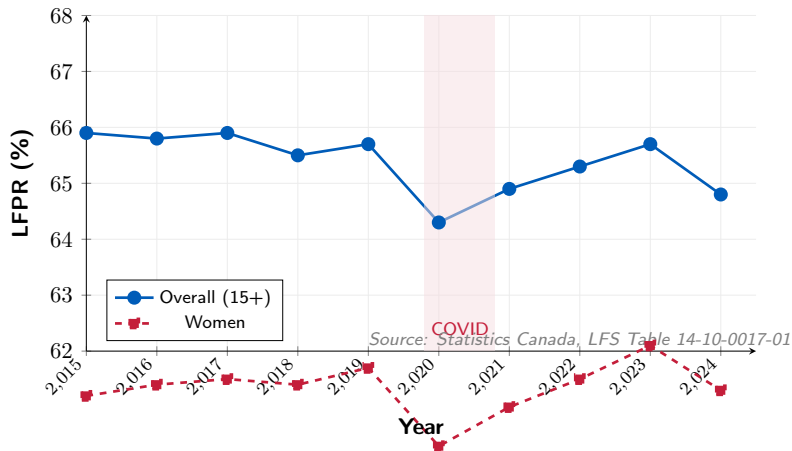
$$LFPR = \frac{\text{Labour force}}{\text{Working-age population}} \times 100$$

- Employment-population ratio (ER):

$$ER = \frac{\text{Number employed}}{\text{Working-age population}} \times 100$$

- LFPR distinguishes between low unemployment because people *have jobs* vs. because they've *given up looking*.

Figure 5.2 – Labour Force Participation Rate: Canada 2015–2024



Problems with Measuring Unemployment

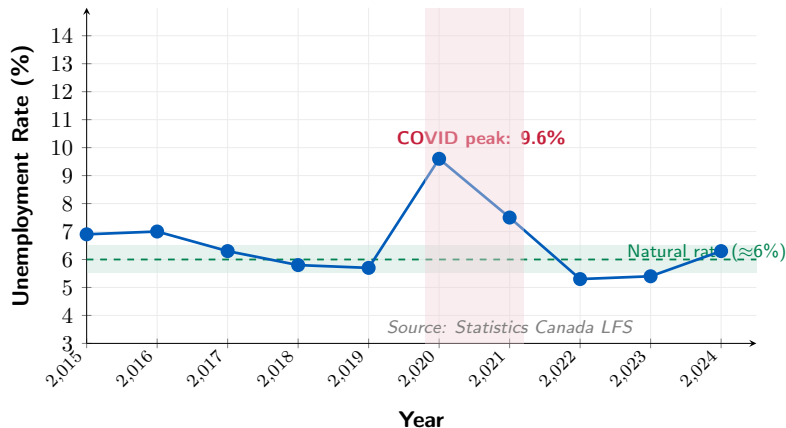
- The measured UR is **not perfect**.
- May **understate** true unemployment:
 - **Discouraged workers:** stopped looking but would accept a job — classified as “not in the labour force.”
 - **Underemployment:** part-time workers who want full-time work; those in jobs below their skill level.
 - Statistics Canada's **R8 rate** includes these: in 2024, $R8 \approx 9.8\%$ vs. official UR of 6.3%.
- May **overstate** unemployment:
 - Some people claim to search while not seriously looking (to retain benefits).

5.2 Learning Objective

Goal

Identify and understand the four types of unemployment.

Figure 5.3 – Canadian Unemployment Rate 2015–2024



Four Types of Unemployment

- **Frictional:** short-term; workers between jobs or entering the market.
- **Structural:** persistent mismatch between worker skills and job requirements.
- **Cyclical:** caused by a recession — demand for labour falls economy-wide.
- **Seasonal:** due to predictable seasonal patterns (weather, harvest, holidays).

- **Frictional unemployment:** short-term unemployment from the process of **matching** workers to jobs.
- Causes:
 - People entering or re-entering the labour force.
 - People voluntarily leaving one job to search for a better one.
- Some frictional unemployment is **unavoidable** and even **beneficial**:
 - better matches $\Rightarrow$ higher productivity.
- **Canadian example:** A recent UBC graduate takes 8 weeks to find an entry-level data analyst role in Vancouver — this is frictional unemployment.

- **Structural unemployment:** persistent mismatch between the **skills** of workers and the **requirements** of available jobs.
- Causes: technological change, industry decline, geographic mismatch.
- Tends to be **longer-term** — workers may need retraining or relocation.
- **Canadian examples:**
 - Auto assembly workers in Windsor displaced by automation.
 - Coal miners in Alberta's Crowsnest Pass as coal demand declines.
 - Ottawa's Better Jobs program targets retraining for structurally unemployed workers.

- **Cyclical unemployment:** unemployment caused by a **business-cycle recession**.
- When aggregate demand falls, firms cut production and lay off workers.
- Workers are skilled and would normally be employed — the problem is insufficient demand.
- **Canadian examples:**
 - COVID-19 (Spring 2020): unemployment surged from 5.7% to 13.7% in two months.
 - 2008–09 Global Financial Crisis: Canadian UR rose from 6% to 8.7%.
 - Chrysler Brampton Assembly (2009): temporary closure due to low car demand.

- **Seasonal unemployment:** predictable unemployment due to weather, calendar, or demand patterns.
- **Canadian examples:**
 - Construction workers in Saskatchewan laid off each November.
 - Ski resort staff in Banff unemployed in summer.
 - Fish plant workers in N.L. idle between fishing seasons.
 - Retail staff hired in November–December, let go in January.
- Statistics Canada publishes both **seasonally adjusted** and **unadjusted** series to separate this effect.

Full Employment and the Natural Rate

- At **full employment**, all unemployment is frictional and structural — zero cyclical unemployment.
- **Natural rate of unemployment**: the unemployment rate when the economy is at full employment.

$$u^* = \text{Frictional} + \text{Structural unemployment}$$

- **Canadian estimates**: $u^* \approx 5.5\text{--}6.0\%$ in recent years (Bank of Canada, MPR October 2024).
- When actual UR $> u^*$: cyclical unemployment exists $\Rightarrow$ output gap is negative.

5.3 Learning Objective

Goal

Explain what determines the unemployment rate and how policy affects it.

- **Employment Insurance (EI):** replaces $\approx 55\%$ of insurable earnings (max $\approx \$695/\text{week}$ in 2024).
- **Benefits of EI:**
 - Gives workers time to find better-matched jobs.
 - Stabilizes consumer spending during recessions.
- **Costs of EI:**
 - Reduces the urgency to find work $\Rightarrow$ **longer average duration** of unemployment.
 - Estimates suggest EI increases frictional unemployment slightly.

Minimum Wage Laws

- Minimum wages raise the cost of hiring labour for firms.
- May cause firms to hire fewer workers, substitute capital for labour, or reduce hours.

Province / Territory	Min. Wage (2025)
Nunavut	\$19.00/hr
British Columbia	\$17.40/hr
Ontario	\$17.20/hr
Alberta	\$15.00/hr
New Brunswick	\$15.30/hr

- Research suggests a 10% increase in minimum wage reduces teenage employment by $\approx 2\%$.
- Overall effect on total unemployment is believed to be **relatively small** (Bank of Canada, 2023).

- **Labour unions:** $\approx 30\%$ of Canadian workers are unionized (2023); $\approx 77\%$ of public sector vs. $\approx 15\%$ private sector.
 - Unions push wages above competitive levels for members $\Rightarrow$ some unemployment for non-members.
 - Limited impact on **overall** unemployment in Canada.
- **Efficiency wages:** firms voluntarily pay above-market wages to:
 - reduce turnover (e.g., large banks, tech firms),
 - increase effort and attract higher-quality workers.
- Result: persistent unemployment even in good times — some workers *want* jobs at the efficiency wage but cannot find them.

5.4 Learning Objective

Goal

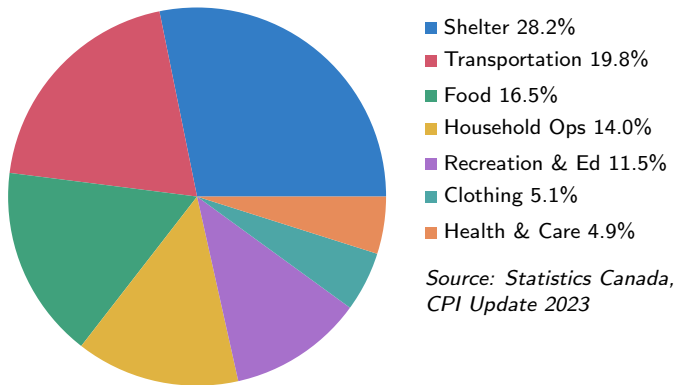
Define the price level and inflation rate, and understand how they are computed.

- **Price level:** average prices of goods and services in the economy.
- **Inflation rate:** percentage increase in the price level from one year to the next.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$$

- Key measures: **CPI**, **PPI**, and **GDP deflator**.

Figure 5.4 – The Canadian CPI Basket (2023 Weights)



The Consumer Price Index (CPI)

- **CPI:** measures the average change over time in the prices paid by a typical Canadian household.
- Based on a fixed **basket of goods** (quantities from base period 2002).
- **Formula:**

$$\text{CPI}_t = \frac{\text{Cost of basket in year } t}{\text{Cost of basket in base year}} \times 100$$

- Used to index wages, pensions, EI benefits, and Old Age Security.
- Bank of Canada's inflation target: **2%** (with a control range of 1–3%).

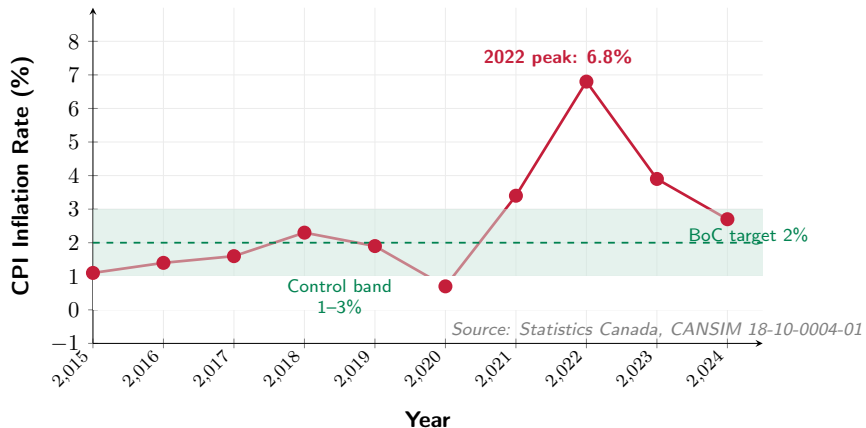
Calculating the CPI: A Canadian Example

- Basket: 5 coffees at Tim Hortons, 1 transit pass, 2 kg of canola oil.

Item	2002 (base)		2024		Note
	Price	Qty	Price	Qty	
Coffee (cup)	\$1.25	5	\$2.50	5	doubled
Transit pass	\$60	1	\$105	1	+75%
Canola oil (kg)	\$2.50	2	\$4.00	2	+60%
Basket cost	\$71.25		\$128.50		

- $$CPI_{2024} = \frac{128.50}{71.25} \times 100 \approx 180.4$$
- This means prices in this basket rose $\approx 80\%$ since 2002.

Figure 5.5 – Canadian CPI Inflation Rate 2015–2024



Is the CPI Accurate? Four Biases

- The CPI tends to **overstate** true inflation by ≈ 0.5 –1 percentage point per year.
- **Substitution bias**: fixed basket ignores consumers switching to cheaper substitutes.
 - If beef prices rise, consumers buy more chicken — but the basket keeps both at old quantities.
- **Quality change bias**: price increases partly reflect better quality (e.g., safer, more fuel-efficient cars).
- **New product bias**: smartphones, streaming services entered the basket with a lag.
- **Outlet bias**: shift to Costco, Amazon, and discount retailers not fully reflected.

Producer Price Index (PPI)

- **PPI:** measures average prices received by **producers** at all stages of production.
- Basket includes raw materials, semi-finished goods, and wholesale prices.
- Changes in the PPI **lead** changes in the CPI:
 - Higher input costs today $\Rightarrow$ higher retail prices tomorrow.
 - **Example:** Surging global wheat prices in 2022 (Russia–Ukraine war) raised PPI for Canadian bakers $\Rightarrow$ grocery bread prices followed within 3–6 months.

5.5 Learning Objective

Goal

Use price indexes to adjust nominal values for inflation.

- To compare purchasing power across years, convert nominal amounts to **real** amounts.

$$\text{Real value}_t = \text{Nominal value}_s \times \frac{\text{CPI}_t}{\text{CPI}_s}$$

- **Example 1 – Minimum wage:**

- Ontario minimum wage was \$10.25/hr in 2014; CPI was 122. In 2024, CPI $\approx$ 165.

$$\text{Real}_{2024} = 10.25 \times \frac{165}{122} \approx \$13.86/\text{hr}$$

- Ontario's actual 2024 minimum of \$17.20/hr represents a genuine real increase.

- **Example 2 – Average weekly earnings:**

- Average weekly earnings 2014: \$953; 2024: \$1,260.
- Nominal increase: +32%. CPI increase 2014–2024: $\approx$ 35%.
- Real wages actually **fell** slightly.

5.6 Learning Objective

Goal

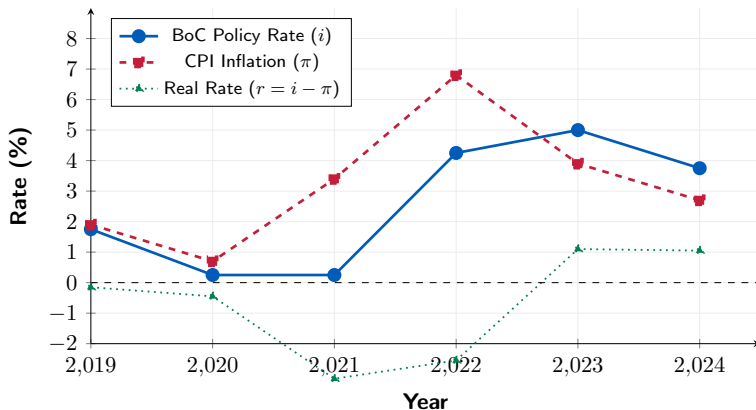
Distinguish between nominal and real interest rates.

- **Nominal interest rate (i):** the stated rate on a loan or deposit.
- **Real interest rate (r):** the nominal rate adjusted for inflation:

$$r \approx i - \pi$$

- The real rate measures the **true purchasing-power gain** from lending (or cost of borrowing).

Figure 5.6 – Bank of Canada Policy Rate vs. CPI Inflation 2019–2024



Source: Bank of Canada; Statistics Canada CANSIM 18-10-0004-01. Approximate annual averages.

Real vs. Nominal Interest Rates: Canadian Context

- 2021: BoC policy rate = 0.25%, inflation = 3.4% $\Rightarrow$ real rate = **-3.15%** — borrowers were rewarded!
- 2023: BoC policy rate = 5.0%, inflation = 3.9% $\Rightarrow$ real rate = **+1.1%** — tighter conditions.
- 2024: BoC began cutting rates as inflation returned toward 2%.
- **Key lesson:** a high nominal rate is not necessarily tight if inflation is also high — what matters for real decisions is the **real rate**.

5.7 Learning Objective

Goal

Discuss the problems inflation can cause, even when anticipated.

Is Inflation a Problem?

- If all prices and wages rose at **exactly the same rate**: relative prices unchanged, purchasing power unchanged — inflation would be harmless.
- In practice:
 - Not all prices adjust simultaneously or equally.
 - Some wages are fixed in long-term contracts.
 - Cash and fixed-income assets lose real value.
- **Canadian 2022 example**: Grocery prices rose $\approx 10\%$ while many workers on annual contracts saw only 3–4% raises — real wages fell.

Costs of Anticipated Inflation

- **Redistribution of income:** workers on fixed contracts lose purchasing power; asset owners with real assets (houses, stocks) gain.
- **Loss of purchasing power of cash:** “shoe-leather costs” — people make more trips to the bank, hold less cash.
- **Menu costs:** firms must update price lists, menus, vending machines — real resource costs. Canada’s restaurant sector spent heavily reprinting menus in 2022.
- **Bracket creep:** nominal income rises with inflation, pushing households into higher tax brackets even if real income is unchanged.
- **Taxation of nominal returns:** investors taxed on nominal capital gains even when the real return is zero.

Costs of Unanticipated Inflation

- **Unanticipated inflation** is especially harmful — it makes long-term contracting difficult.
- **Borrowers vs. lenders:**
 - Higher-than-expected inflation: borrowers **gain** (repay in cheaper dollars); lenders **lose**.
 - Lower-than-expected inflation: borrowers **lose**; lenders **gain**.
- **Canadian 1980s example:** Mortgage rates reached 18–21%; homeowners who locked in suffered when inflation and rates fell later.
- **2021–22 example:** Many fixed-rate mortgage holders are now renewing at rates 3–4× higher — an unanticipated real cost.
- Unpredictable inflation discourages saving, investment, and long-term planning.

Things to Remember

- The unemployment rate, LFPR, and ER together describe labour market health — no single number tells the whole story.
- Four types of unemployment: frictional (unavoidable), structural (skill mismatch), cyclical (recession), seasonal (weather/calendar).
- CPI measures cost of living; it has biases that cause it to **overstate** inflation by $\approx 0.5\text{--}1\%$.
- Real values = nominal values adjusted by a price index — always use real for welfare comparisons.
- Real interest rate $\approx$ nominal rate — inflation rate — what matters for borrowing and saving decisions.
- Inflation, especially unanticipated, redistributes income and distorts economic decisions.

- Statistics Canada measures labour market via LFS ($\approx 56,000$ households/month).
- Natural rate $u^* \approx 5.5\text{--}6\%$; cyclical UR in 2020 peaked at 13.7% (COVID).
- Bank of Canada targets 2% CPI inflation; in 2022 CPI hit 6.8% — highest since 1982.
- CPI basket: shelter (28%), transport (20%), food (17%) are the largest components.
- Real interest rate in 2021: $\approx -3\%$; in 2023: $\approx +1\%$ — dramatic shift in monetary conditions.
- Unanticipated inflation creates winners (borrowers) and losers (lenders, fixed-income holders).

Thank you!

Questions?

Data: Statistics Canada LFS (14-10-0017-01), CPI (18-10-0004-01), National Accounts; Bank of Canada MPR 2024